

COLLEGE PARK, MD, USA

WMO: 722244

Lat: 38.981N

Lon: 76.922W

Elev: 50

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 00358

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 14.9	(c) 19.1	(d) -3.7	(e) 4.5	(f) 20.0	(g) 1.1	(h) 5.8	(i) 25.0	(j) 19.9	(k) 35.5	(l) 18.2	(m) 34.7	(n) 5.3	(o) 240	(p) 0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	95.2	74.2	92.0	73.6	90.2	73.0	78.0	87.5	76.9	86.2	75.7	85.0	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	133.7	80.7	74.2	127.7	79.8	73.0	122.9	79.2	41.3	88.6	40.1	86.4	39.1	84.7	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.7	(b) 13.7	(c) 11.5	(e) 9.4	(f) 98.5	(g) 3.3	(h) 3.6	(i) 7.0	(j) 101.1	(k) 5.1	(l) 103.2	(m) 3.3	(n) 105.2	(o) 0.8	(p) 107.8		
			DB													
			WB	8.3	79.6	3.6	1.4	5.7	80.5	3.6	81.3	1.6	82.1	-1.0	83.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.5	35.4	38.0	45.3	56.8	65.8	74.5	78.8	76.9	70.2	58.9	47.2	40.6
	DBStd	17.02	9.14	9.48	9.20	8.71	7.80	5.82	5.21	4.85	6.48	8.22	7.77	8.25
	HDD50	1526	463	353	206	30	0	0	0	0	0	15	144	315
	HDD65	4197	918	756	613	275	91	4	0	0	22	226	535	757
	CDD50	4253	10	17	60	233	490	735	894	833	607	292	59	23
	CDD65	1450	0	0	2	29	115	290	429	368	179	38	0	0
	CDH74	14653	0	3	32	417	1136	2952	4725	3552	1532	298	6	0
	CDH80	5920	0	0	2	136	366	1176	2181	1455	535	69	0	0
(14) Wind	WSAvg	3.6	4.5	4.7	4.9	4.6	3.4	3.2	2.8	2.4	2.7	3.1	3.5	3.7
(15) Precipitation	PrecAvg	42.7	3.1	2.8	3.6	3.1	4.3	3.5	4.1	4.4	3.8	3.2	3.4	3.4
	PrecMax	59.2	8.3	6.9	8.0	7.5	9.1	11.8	10.8	11.0	13.8	8.0	7.2	7.1
	PrecMin	29.4	0.6	0.6	0.7	0.9	1.1	0.6	0.7	0.2	0.4	0.0	0.5	0.6
	PrecStd	7.1	1.6	1.6	1.7	1.6	1.9	2.2	2.2	2.7	3.0	1.9	1.9	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.7	72.3	78.9	90.2	90.8	97.3	99.9	96.8	93.2	87.6	73.4	70.1
		MCWB	60.7	59.8	59.1	64.8	71.9	73.9	74.2	74.7	72.9	71.3	60.4	63.6
	2%	DB	61.4	64.3	72.5	83.8	87.9	92.7	96.6	93.0	90.3	81.1	69.0	64.0
		MCWB	55.1	54.9	58.8	64.9	70.9	72.9	74.0	74.0	72.7	68.4	59.2	60.7
	5%	DB	55.4	59.3	67.7	79.0	84.4	90.1	93.1	90.3	86.4	77.1	64.2	59.2
		MCWB	48.2	49.2	56.7	62.6	69.3	72.4	74.7	73.6	71.3	66.2	56.2	53.9
	10%	DB	50.1	53.1	62.5	73.1	81.1	87.5	90.4	88.0	82.2	72.8	61.0	53.9
		MCWB	44.4	44.6	51.8	59.8	67.2	71.4	73.7	72.7	69.4	64.3	53.6	48.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.7	62.3	63.2	70.2	74.8	77.9	79.9	79.7	77.1	74.0	64.8	65.5
		MCDB	64.5	67.8	69.9	79.7	85.1	87.3	90.5	90.1	85.1	79.8	69.6	67.8
	2%	WB	57.2	57.9	60.6	67.2	72.8	76.3	78.2	77.8	75.5	71.9	61.8	61.6
		MCDB	61.4	65.0	69.4	76.8	83.0	86.9	88.3	86.3	84.2	78.0	66.3	65.1
	5%	WB	49.9	50.2	57.4	64.6	71.4	75.0	77.1	76.4	74.2	69.8	58.6	55.4
		MCDB	54.3	56.5	66.8	76.1	80.9	85.4	87.1	84.6	81.9	75.1	63.5	58.7
	10%	WB	44.7	46.0	53.3	61.9	69.9	73.7	76.0	75.2	72.7	66.5	55.1	50.2
		MCDB	48.6	51.9	59.8	72.4	77.7	82.9	85.6	83.3	79.6	71.8	59.9	52.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.0	19.0	21.7	20.2	20.0	19.3	18.8	19.3	19.8	18.7	15.3
		MCDBR	24.4	26.5	27.6	29.6	25.4	23.2	22.3	21.7	23.5	23.4	24.7	21.1
		MCWBR	18.9	17.3	16.5	12.7	10.8	7.6	6.2	7.0	7.9	11.6	15.1	17.6
	5% WB	MCDBR	21.7	23.1	25.2	25.6	21.9	19.8	18.7	18.3	18.0	18.8	18.8	18.9
		MCWBR	19.1	18.1	17.5	13.2	10.1	7.7	6.3	6.9	7.3	11.5	14.7	17.4
(40) Clear-Sky Solar Irradiance	taub		0.310	0.323	0.358	0.403	0.444	0.501	0.521	0.516	0.430	0.363	0.325	0.311
	taud		2.484	2.429	2.396	2.290	2.205	2.088	2.047	2.046	2.262	2.452	2.517	2.518
	Ebn at Noon		275	286	285	278	267	251	244	242	258	267	267	266
	Edh at Noon		25	30	35	41	45	51	53	52	39	29	24	23
(44) All-Sky Solar Radiation	RadAvg		656	925	1229	1561	1756	1943	1909	1699	1384	1034	748	575
	RadStd		54	104	89	120	200	148	97	92	162	114	63	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+223	+128

Nomenclature: See separate page